

Why the Current Reconstruction Error Plateaus

A diagnostic note on fixed-kernel looped attention

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Abstract

The present proof of concept does show that prescribing the nonlinear softmax/RBF feature geometry is useful: the matched loop outperforms both a standard Transformer and the same loop with a mismatched kernel. It does *not*, however, yet produce uniformly accurate reconstructions. The reason is primarily informational rather than an optimization failure. With 12 random measurements, the trained loop has MSE 0.12055, while the exact Gaussian-process posterior mean—using the true kernel and noise level—is already limited to 0.11359. Thus 94.2% of the observed error remains even after replacing the learned iteration by an exact solve. At budget 8, the corresponding fraction is 98.0%. A second issue is objective misalignment: experimental design was trained with a spatial weight that is 5.78 times larger at the centre than at the boundary, so a low weighted loss can coexist with visibly poor boundary reconstruction. Analytic posterior-risk calculations show that roughly 16 well-placed measurements, rather than 8, are needed to cross MSE 10^{-2} in this one-dimensional problem. More training alone cannot remove this floor.

1 Exact decomposition of the prediction error

Let $f \sim \mathcal{N}(0, K)$ on the fixed grid and $y_C = f_C + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. For a spatial loss matrix $W = \text{diag}(w)$, the exact posterior mean and covariance are

$$m_C = K_{:C}(K_{CC} + \sigma^2 I)^{-1} y_C, \quad (1)$$

$$\Sigma_C = K - K_{:C}(K_{CC} + \sigma^2 I)^{-1} K_{C:}. \quad (2)$$

For any predictor $\hat{f}(y_C)$, posterior orthogonality gives

$$\underbrace{\mathbb{E}\|\hat{f} - f\|_W^2}_{\text{total risk}} = \underbrace{\mathbb{E}_C \text{tr}(W \Sigma_C)}_{\text{information and noise floor}} + \underbrace{\mathbb{E}\|\hat{f} - m_C\|_W^2}_{\text{solver/model excess}}. \quad (3)$$

Because the data generator uses the same RBF kernel, $\ell = 0.18$, and $\sigma = 0.1$, exact KRR with ridge $\lambda = \sigma^2 = 0.01$ is the Bayes posterior mean. It is therefore the correct diagnostic floor for this protocol, not merely another baseline.

Table 1: Empirical risk decomposition over five independent training seeds.

Protocol	loop risk	exact GP risk	loop excess	floor/total
12 random sensors, uniform loss	0.12055	0.11359	0.00696	94.2%
8 learned sensors, weighted loss	0.06049	0.05929	0.00120	98.0%

The loop is therefore not the dominant bottleneck. Increasing the tied depth from 12 to 24 reduces the random-context MSE from approximately 0.1201 to 0.1146, already close to the exact value 0.1136. Additional optimizer steps can at most affect the small second term of Eq. (3); they cannot reduce posterior uncertainty.

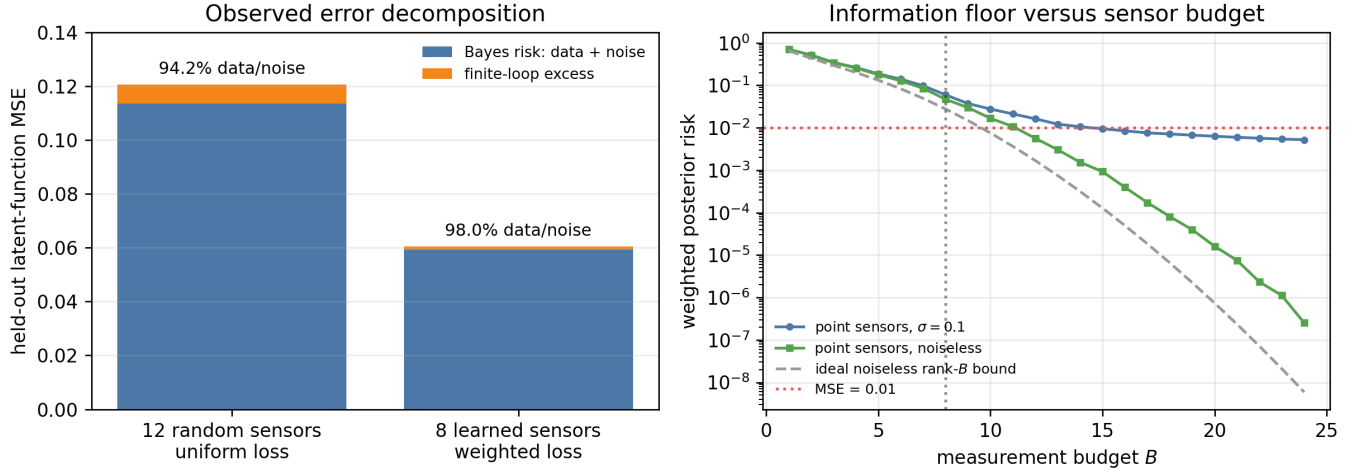


Figure 1: **Left:** almost all current error is the exact Bayesian risk conditioned on the available measurements. **Right:** analytic weighted posterior risk for sequential point measurements, with and without observation noise. The grey curve is the more optimistic spectral lower bound obtained if one were allowed arbitrary noiseless rank- B linear measurements rather than point sensors. The current budget $B = 8$ is marked vertically.

2 The primary bottleneck is the measurement budget

At budget 8, three effects can be separated analytically for the weighted design objective:

Quantity	weighted posterior risk
Ideal noiseless rank-8 linear-information bound	0.02765
Greedy point design, noiseless	0.04669
Greedy point design, $\sigma = 0.1$	0.05950
Trained point design + exact GP, $\sigma = 0.1$	0.05929
Trained point design + loop, $\sigma = 0.1$	0.06049

Even an unrealistically favourable noiseless linear sensing operator cannot make the rank-8 risk much smaller than 2.8×10^{-2} under this prior. Restricting observations to locations raises the risk further, and noise adds another 1.28×10^{-2} for the greedy point design. By contrast, the finite loop contributes only 1.20×10^{-3} for the trained design.

The regime changes near 16 well-placed sensors. For the same weighted objective, greedy posterior risks are

B	8	10	12	14	16	24
$\sigma = 0.1$	.05950	.02760	.01624	.01056	.00846	.00522
noiseless	.04669	.01670	.00566	.00154	.000395	$< 10^{-7}$

Thus $B \simeq 16$ is the first sensible operating point for an MSE target below 10^{-2} . Once the domain is adequately covered, observation noise becomes the leading term; repeated measurements or lower-noise sensors then matter more than additional training.

3 The reported design loss was not a global reconstruction loss

The design experiment minimized

$$\mathcal{R}_w(\hat{f}) = \sum_{q=1}^{64} w(x_q) (\hat{f}(x_q) - f(x_q))^2, \quad w(x) \propto 0.2 + \exp\left[-\frac{x^2}{2(0.32)^2}\right].$$

Consequently $w(0)/w(\pm 1) = 5.78$. Boundary errors are deliberately cheap. This was useful for testing whether a policy could adapt to a nonuniform query distribution, but it was the wrong objective for demonstrating visually good reconstruction over the full interval. The previous qualitative panel also showed one arbitrary draw without uncertainty, which made the numerical result hard to interpret.

Figure 2 uses a held-out episode selected as the median exact-GP weighted error among 512 fresh draws, rather than selecting an easy example. With the current learned $B = 8$ design, the global MSE is 0.145 even though the weighted MSE is 0.061. Running the same loop for 32 iterations makes it coincide with the exact posterior mean but leaves the global MSE at 0.140. Changing the information set to 16 globally placed measurements reduces it to 0.010. The posterior band makes the missing information visible and distinguishes uncertainty from solver error.

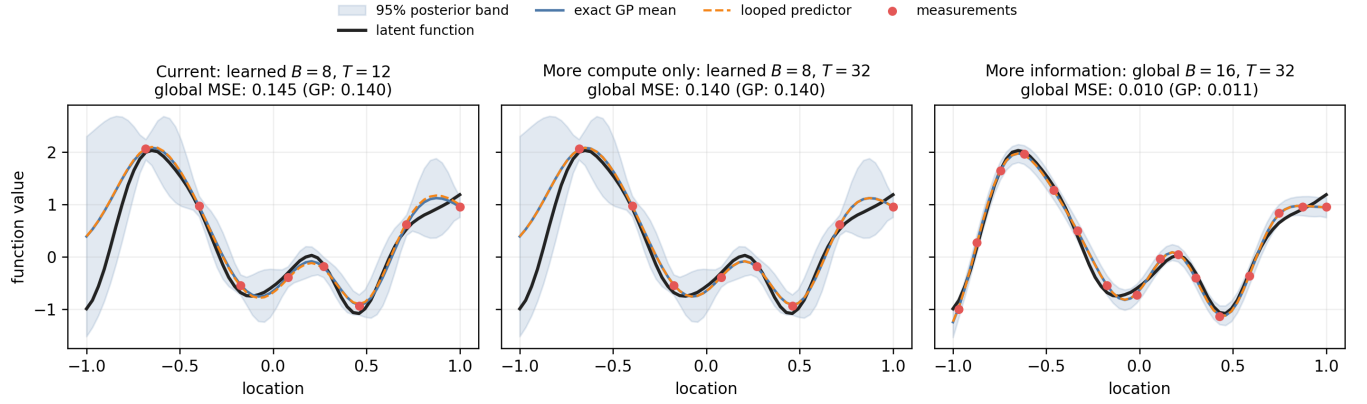


Figure 2: A representative held-out function. More recurrent computation aligns the loop with the exact GP mean but cannot recover the unobserved left boundary. Doubling the budget and optimizing global coverage removes the dominant uncertainty and produces a genuinely accurate reconstruction.

4 What should change in the next experiment

If the target is small *global* reconstruction error, the next protocol should make the following changes.

1. Replace the centre-weighted loss by uniform integrated MSE, or by a robust combination of integrated and worst-location posterior risk.
2. Train and evaluate designs at budgets $B \in \{12, 16, 20, 24\}$; retain $B = 8$ only as a deliberately underdetermined stress test.
3. Use 24–32 tied iterations at inference. This removes nearly all of the solver excess without changing the fixed nonlinear feature geometry.
4. Report exact-GP risk, loop excess, global MSE, weighted MSE, and posterior calibration separately. A single loss number is insufficient.
5. For the fixed LSM/PDE geometry, compute the same posterior trace before training. This determines whether the proposed sensor budget can possibly attain the desired tolerance and prevents spending training time below an information-theoretic floor.

The modelling conclusion about the nonlinearity remains intact: choosing the softmax feature map to match the kernel is consequential. At 12 observations, the matched loop has MSE 0.1205, compared with 0.2090 for the same loop with length scale $\ell/2$ and 0.1370 for the standard Transformer. What that choice cannot do is manufacture information absent from the measurements.

5 Scope

These numbers concern a finite 64-point, one-dimensional RBF GP with fixed geometry. The spectral curve is a model-conditioned Bayesian bound, not a universal scaling law, and the experiment is not yet evidence for a PDE inverse problem. Its useful conclusion is narrower and operational: the present reconstruction plateau is caused chiefly by an undersized and misaligned measurement objective, while the learned loop is already close to the exact kernel regressor.